Arman Motiwala

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EDUCATION

Northeastern University, Khoury College of Computer Sciences **Sep 2025 - Dec 2027**
Master of Science, Artificial Intelligence *Boston, MA*
- **Coursework:** Algorithms, Foundations of AI, Machine Learning and Pattern Recognition, Reinforcement Learning, NLP

Mukesh Patel School of Technology Management and Engineering **Jul 2021 - May 2025**
Bachelor of Technology, Artificial Intelligence *Mumbai, India*
- **Coursework:** Machine Learning, Deep Learning, NLP, Advanced ML, Recommendation Systems, Operating Systems, Blockchain Technology, Computer Vision

SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Java
ML & GenAI: PyTorch, scikit-learn, Transformers, LoRA, Reinforcement Learning, RAG, LangGraph, LlamaIndex, Regression Analysis, Statistics, Computer Vision, NLP, Transformers, Large Language Models, vLLM, CNN, RNN, LSTM, LLM-Finetuning, Reinforcement Learning, Time Series Analysis, LlamaIndex, Ollama
Engineering: FastAPI, Next.js, Docker, PostgreSQL, Qdrant, Milvus, GitHub Actions, GCP/Vertex AI, AWS, OpenAI API, PySpark, Dokploy, React, Ablation Studies, Benchmark Design, Applied Statistics, Kubernetes, Backend Development, API Integration, Cloud Deployment, Vector Databases

EXPERIENCE

Northeastern University, D'Amore-McKim AI Strategic Hub | *AI Engineer* **May 2026 - Present**
- Build a full-stack AI-TA platform with backend API integrations that delivers sub-2-second, RAG-grounded chat responses across 50+ course documents using Next.js 15, TypeScript, FastAPI, Qdrant, and LlamaIndex.
- Automate Canvas LMS synchronization and grading workflows for 6+ courses through a dual-deployment CI/CD pipeline, integrating GitHub Actions, Qdrant, and PostgreSQL.
- Develop Impact Connect, a role-based engagement platform connecting DMSB alumni and executive leaders with faculty and students across 5 user roles, using Next.js, TypeScript, PostgreSQL, and Better Auth, with admin approval workflows and fit-score search.
- Collaborate on MatchIQ, a human-in-the-loop record-matching pipeline linking 68K+ alumni records to RocketReach profiles through confidence-scored name, location, and degree matching, using Google Apps Script and the RocketReach API for shared review and live data enrichment.

Know Loctoc | *Software Developer Intern* **Jan 2025 - Jul 2025**
- Achieved 95% visual-text alignment accuracy across 1,000+ product images with Vertex AI and GPT-4o; cut manual audit workload 90% through a FastAPI/Milvus NLP feedback pipeline (F1: 0.87; ROC-AUC: 0.91).
- Designed a RAG-based adaptive quiz system using OpenAI LLMs and vector search, generating 6 Bloom's Taxonomy-aligned question types with adversarial robustness testing.

Sequel String AI | *AI/ML Intern* **Apr 2024 - Jul 2024**
- Boosted structured-data throughput 40% by deploying an OpenCV and REST API pipeline that processed 500+ utility-meter images daily.
- Extended multilingual accessibility across 3 regional languages by integrating neural machine translation and language-detection modules.

PROJECTS

Human-in-the-Loop Reinforcement Learning with Preference-Based Rewards **Jan 2026 - May 2026**
- Architected and trained a PyTorch reward model on 1,200+ pairwise human preferences using Bradley-Terry comparisons, then evaluated policy alignment across 4 MuJoCo tasks, improving performance 31% over baseline PPO.
- Conducted annotation-budget and feedback-noise ablations to quantify tradeoffs, reducing required human queries 40% while retaining 94% of full-feedback policy performance.

Medical Image Analysis | Github **Oct 2025 - Dec 2025**
- Developed and evaluated X-ray models for fracture segmentation using U-Net++/EfficientNet-B4, achieving 61.48% Dice and 93.14% classification accuracy, and chest X-ray classification using EfficientNet-B3, achieving 84.92% accuracy and 0.84 macro-F1.
- Addressed extreme class imbalance with a custom Dice-Focal-BCE loss and implemented and tested both deep learning models in PyTorch.

PUBLICATIONS

Auto-Assess: Automated Evaluation Platform for Enhanced SQL Learning (IEEE SCEEECS 2025). Read here
- Fine-tuned Llama 3.1 with LoRA on Spider, attaining 87% execution accuracy on WikiSQL and publishing an automated SQL-evaluation framework at IEEE SCEECS 2025.
- Built an evaluation pipeline across 8,000+ queries for execution accuracy and logical-form correctness using Hugging Face Transformers, PyTorch, MongoDB, and Flask.